FarmConnect Technology Stack Speech

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Good [morning/afternoon], everyone. This section explains the technologies used in the FarmConnect project and why each one was chosen.

FarmConnect is built as a monorepo, which allows the web app, API, and AI service to share standards and evolve together. The workspace is managed with pnpm and Turborepo for fast, repeatable builds.

On the frontend, we use React with TypeScript and Vite. React provides a modular component system, TypeScript adds type safety, and Vite keeps development fast. The UI is styled with Tailwind CSS, and the app supports multiple languages including English and Arabic.

The backend API is built with Node.js, Express, and TypeScript. This layer handles authentication, business logic, and role-based access. Database access is managed with Prisma, which provides a clean schema and type-safe queries. PostgreSQL is used as the relational database.

For AI features, we use a separate Python FastAPI service. This service exposes endpoints for price recommendation, demand forecasting, and chatbot assistance. Keeping AI in its own service isolates heavy computation and allows independent scaling.

Shared code is organized into internal packages. The shared package contains common enums and types used across the API and web app. The UI package provides reusable components. The config package centralizes linting, formatting, and TypeScript settings.

For infrastructure and deployment, the project defines environment variables for each app. The API, web, and AI services can be deployed independently. Docker is used for the AI service containerization.

Testing spans both frontend and AI service components. The web app uses Vitest for unit and component tests. The AI service includes pytest-based tests for logic and endpoints.

In summary, the stack combines a modern React frontend, a TypeScript API with Prisma and PostgreSQL, and a Python FastAPI AI layer. This balance supports rapid development, strong type safety, and scalable services. Thank you.